

Problem Sheet: Functions 1 Solution

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Exercise 1. Let f be a function defined by

$$f(x) = \frac{3x+2}{7}.$$

1. Find $ff(x)$, and write your answer in the form

$$ff(x) = \frac{ax+b}{49}$$

where a, b are numbers.

2. Find $f^{-1}(x)$, and write your answer in the form

$$f^{-1}(x) = \frac{ax+b}{3}$$

where a, b are numbers.

Solution. 1. First write

$$f(x) = \frac{3x+2}{7}.$$

Then

$$ff(x) = f(f(x)) = \frac{3f(x)+2}{7}.$$

Substituting $f(x)$ in gives

$$ff(x) = \frac{3\frac{3x+2}{7}+2}{7}.$$

Multiply by $\frac{7}{7}$ to remove the inner denominator:

$$\begin{aligned} ff(x) &= \frac{7}{7} \times \frac{3\frac{3x+2}{7}+2}{7} \\ &= \frac{3 \times \cancel{7} \times \frac{3x+2}{\cancel{7}} + 14}{49} \\ &= \frac{9x+6+14}{49} \\ &= \frac{9x+20}{49}. \end{aligned}$$

Hence

$$ff(x) = \frac{9x+20}{49}$$

and so $a = 9$, $b = 20$.

2. Write

$$y = \frac{3x+2}{7}.$$

Rearranging for x :

$$\begin{aligned} 7y &= 3x + 2 \\ 7y - 2 &= 3x \\ x &= \frac{7y - 2}{3}. \end{aligned}$$

Therefore

$$f^{-1}(x) = \frac{7x - 2}{3}.$$

Hence $a = 7$, $b = -2$.

□

Exercise 2. Let f be a function defined by

$$f(x) = \frac{3x + 2}{6x}, \quad x \neq 0.$$

1. Find $ff(x)$, simplifying your answer into the form

$$ff(x) = \frac{ax + b}{cx + d}$$

where a, b, c, d are numbers.

2. State where $ff(x)$ is defined.
3. Find $f^{-1}(x)$ in the form

$$f^{-1}(x) = \frac{1}{ax + b}$$

where a, b are numbers.

4. State where $f^{-1}(x)$ is defined.

Solution. 1. First write

$$f(x) = \frac{3x + 2}{6x}, \quad x \neq 0.$$

Then

$$ff(x) = f(f(x)) = \frac{3f(x) + 2}{6f(x)}.$$

Substituting gives

$$ff(x) = \frac{3 \frac{3x+2}{6x} + 2}{6 \frac{3x+2}{6x}}.$$

Multiply by $\frac{6x}{6x}$:

$$\begin{aligned} ff(x) &= \frac{6x}{6x} \times \frac{3 \frac{3x+2}{6x} + 2}{6 \frac{3x+2}{6x}} \\ &= \frac{3 \times \cancel{6x} \times \frac{3x+2}{\cancel{6x}} + 12x}{6 \times \cancel{6x} \times \frac{3x+2}{\cancel{6x}}} \\ &= \frac{9x + 6 + 12x}{18x + 12} \\ &= \frac{21x + 6}{18x + 12} \\ &= \frac{7x + 2}{6x + 4}. \end{aligned}$$

Hence

$$ff(x) = \frac{7x + 2}{6x + 4}$$

so $a = 7$, $b = 2$, $c = 6$, $d = 4$.

2. We need

$$6x + 4 \neq 0$$

so

$$x \neq -\frac{2}{3}.$$

But we also need $x \neq 0$ from the original function. There is one more thing to check: since $ff(x) = f(f(x))$, we also need

$$f(x) \neq 0.$$

That would mean

$$\frac{3x+2}{6x} = 0,$$

but this has no solution, since a fraction is only 0 when its numerator is 0, and $3x+2=0$ would give $x = -\frac{2}{3}$, which we have already excluded from the denominator of $ff(x)$. Therefore $ff(x)$ is defined whenever

$$x \neq 0, \quad x \neq -\frac{2}{3}.$$

3. Write

$$y = \frac{3x+2}{6x}.$$

Rearranging for x :

$$6xy = 3x + 2$$

$$6xy - 3x = 2$$

$$x(6y - 3) = 2$$

$$\begin{aligned} x &= \frac{2}{6y-3} \\ &= \frac{1}{3y-\frac{3}{2}}. \end{aligned}$$

Therefore

$$f^{-1}(x) = \frac{1}{3x - \frac{3}{2}}.$$

Hence $a = 3$, $b = -\frac{3}{2}$.

4. We need

$$3x - \frac{3}{2} \neq 0$$

and therefore

$$x \neq \frac{1}{2}.$$

□

Exercise 3. Let f and g be functions defined by

$$f(x) = \sin(x) + \cos(2x)$$

$$g(x) = \frac{x-1}{2}.$$

1. Find $fg(x)$, simplifying your answer.

2. Let $h(x) = \frac{3x+2}{7}$ and $k(x) = \frac{7x-2}{3}$. By computing $hk(x)$, verify that k is the inverse of h .

Solution. 1. First write

$$f(x) = \sin(x) + \cos(2x), \quad g(x) = \frac{x-1}{2}.$$

Then

$$fg(x) = f(g(x)) = \sin(g(x)) + \cos(2g(x)).$$

Substituting $g(x)$ gives

$$\begin{aligned} fg(x) &= \sin\left(\frac{x-1}{2}\right) + \cos\left(2 \times \frac{x-1}{2}\right) \\ &= \sin\left(\frac{x-1}{2}\right) + \cos(x-1). \end{aligned}$$

2. First write

$$h(x) = \frac{3x+2}{7}, \quad k(x) = \frac{7x-2}{3}.$$

Then

$$\begin{aligned} h(k(x)) &= \frac{3k(x)+2}{7} \\ &= \frac{3\frac{7x-2}{3}+2}{7} \\ &= \frac{\cancel{3}\frac{7x-2}{\cancel{3}}+2}{7} \\ &= \frac{7x-2+2}{7} \\ &= \frac{7x}{7} \\ &= x. \end{aligned}$$

This already shows the key inverse relation. For completeness, we can also check the other composition:

$$\begin{aligned} k(h(x)) &= \frac{7h(x)-2}{3} \\ &= \frac{7\left(\frac{3x+2}{7}\right)-2}{3} \\ &= \frac{3x+2-2}{3} \\ &= \frac{3x}{3} \\ &= x. \end{aligned}$$

Therefore both $h(k(x)) = x$ and $k(h(x)) = x$, which verifies that k is the inverse of h .

□